

Notes on Kopytov et al. (2024)

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Motivation Supply chain disruptions affect aggregates. Firms can mitigate by switching suppliers.

Research Question How does the firm mitigation behavior affect an economy's production network and macroeconomic aggregates?

Results 1) Higher uncertainty leads producers to purchase from more stable suppliers despite potentially higher prices. The resulting reorganization of the network tends to reduce macroeconomic volatility, but at the cost of a decline in aggregate output. 2) More productive and stable firms have higher Domar weights in equilibrium.

Model Multi-sector (n) economy. Each sector populated by representative firm subject to sector-specific productivity shocks.

- (a) Firm. Firms choose production techniques before shocks are realized. Firms have access to production technology set $\mathcal{A}_i$, where a technology $\alpha_i \in \mathcal{A}_i$ specifies the input used. The production function is

$$F(\alpha_i, L_i, X_i) = e^{\epsilon_i} A_i(\alpha_i) \zeta(\alpha_i) L_i^{1-\sum_{j=1}^n \alpha_{ij}} \prod_{j=1}^n X_{ij}^{\alpha_{ij}}$$

where α_{ij} is the input share of good j used by firm i , ϵ_i is the stochastic sector TFP, $A_i(\alpha_i)$ is the productivity shifter, and $\zeta(\alpha_i)$ is the normalization factor. Uncertainty is captured by ϵ_i .

- (b) Household. Standard CRRA utility with CES consumption aggregator. Seems no labour market here because labour does not enter the utility function. But that's not important perhaps for the model (?).
- (c) Timing. Firms have two stages. In the first stage, firms choose production technology before the shock is realized. In the second stage, labour and intermediate inputs (and household consumption) are chosen, after the realization of ϵ .

The paper first solves for the optimal inputs with unit production. Then the technology problem. Since the paper focuses mainly on production network formation and aggregate fluctuation instead of monetary implications, it assumes competitive market so that prices are equal to unit cost (marginal cost). That is, there is no optimal pricing problem.

- (d) Technology choice. Use SDF to discount. Multiply the profit wedge to the demand. Goods market clearing condition is such that

$$C_i^* + \sum_{j=1}^n X_{ji}^* = Q_i^* = F_i(\alpha_i^*, L_i^*, X_i^*). \quad (0.1)$$

References

Kopytov, Alexandr, Bineet Mishra, Kristoffer Nimark, and Mathieu Taschereau-Dumouchel, “Endogenous Production Networks Under Supply Chain Uncertainty,” *Econometrica*, 2024, 92 (5), 1621–1659.